ZERO DCC

DIGITAL COMMAND AND CONTROL INTERFACE
FOR THE HORNBY ZERO 1 TRAIN CONTROLLER

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The Zero DCC module provides a significant upgrade to Zero 1 users. To use this interface effectively, you must have a Zero 1 Master Controller and DCC equipped and programmed locomotives and points/turnouts.

The interface not only provides an upgrade allowing the use of your existing Zero 1 functions to command a modern DCC system, but it also provides additional DCC functions, not previously available using Zero 1.

Warning: Zero 1 equipped locomotives and accessory modules will likely be damaged using this equipment. The only exception is ZTC decoders that are both Zero 1 and DCC compatible.

DO NOT USE ZERO 1 EQUIPPED LOCOMOTIVES ON TRACKS CONNECTED TO THIS INTERFACE. YOU HAVE BEEN WARNED!!!!

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Commands

All the usual Zero 1 commands should work as before.

- Locomotive select
- Inertia select
- Slave select
- Throttle control
- Forward/Reverse
- Turnouts 1-89 direction (NB: 90-99 are used for new features)
- Panic (NB: Panic only puts all locomotives to zero speed – the track is still live!)

The Zero DCC interface provides the following additional features:

- Large LCD screen to display your locomotive roster, speed and direction
- Control track power, on/off etc. (Accessory 90)
- Program DCC decoder ID (CV1) on a separate program track (Accessory 91)
- Display a track schematic with live point and signal information (Accessory 93)
- Record and replay a Zero 1 session (Accessory 95)
- Selection of up to 16 DCC locomotives for control – can be any 16 from your DCC roster! (Accessory 96)
- Save and recall different rosters of 16 locomotives on built in SD card (Accessory 97)
- Simple head light and whistle/horn control (Accessory 98)
- Separate full control of the DCC 29 CAB functions (Accessory 99)
- USB connection for computer access via JMRI and other software (to DCC-EX)
- WiFi connection for computer or mobile access (to DCC-EX)

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Wiring

The Zero DCC interface box has several power inlets and outlets.

Interface power

You will require the usual power requirements of the Zero 1 Phase 1 & 2 Master Controller and Phase 3 Micromimic Console.

In addition to the Zero 1 power, the interface requires two power sources and access to the Zero 1 track power (inputs):

Inputs	**Comment**
USB connector	This is for USB connection of the DCC-EX module to a computer for JRMI control and programming
12V DC via 5.2 plug	This is to provide module and track power (can be up to 5A)
Zero 1 track power (22V AC)	This is the Zero 1 input signal
Outputs	**Comment**
12V DCC to Track A (main track by default)	This provides the DCC signal to the main track to control your trains.
12V DCC to Track B (programming track)	This provides low current DCC to a special programming track to allow you to set your locomotive number or program via JMRI
Switched Zero 1 track power (22A AC)	This is provided for those using the Zero 1 Phase 3 Micromimic system. This is a direct feed from the Zero 1 Master Controller but is switched so that the additional functions of the Zero DCC interface don't interfere with the Micromimic's panel display output.

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Track Power (Accessory 90)

While the Zero 1 green power light will come on when the Zero 1 has mains power available, that does not mean that the track will be getting power from the Zero DCC interface.

The track settings and status can be seen on the top of the Zero DCC LCD screen.

Function 90 provides access to the track power functions.

To enter function 90, select '9 - 0 - →'
To exit function 90, select '9 - 0 - ←'

By default, the track power functions are set to:

Track A:	Main	(ie main track running)
Track B:	Program	(ie second track for DCC programming)
Power:	Off	

WARNING: Before turning track power on, please ensure that any locomotives and other powered items are suitable for the settings selected (DCC by default).

Options under function 90:

Option 1: Turn track power on: select "1- →'
Option 2: Track A = DCC Main: Track B = DCC Program (default)
Option 3: Track A = DCC Main; Track B = 12V DC CAB No. 999
Option 4: Track A = 12VDC CAB No. 999; Track B = 12V DC CAB No. 998

NOTES:

- When the relay switch is fitted and wired correctly, the ZeroDCC will only provide the Zero 1 signal feed when the Track Power function is switched on (this is to avoid confusion between MicroMimic, real accessories and DCC-EX on the status of accessories)
- Locomotive and accessory data on screen will not change when Track Power is OFF (also to avoid confusion around status).
- Whenever the track settings (ie Option 2 – 4) are selected, the track power will be turned off to protect any locomotives you have on the track that do not match the setting chosen.
- When Option 1 (power on) is selected, the display will show 'TRACK POWER ON' with on a black background, but will only be executed when Accessory is closed (ie '9 - 0 - ←') in which case both DCC power and Zero 1 accessory power will be turned on.

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Panic!

Zero 1's old Panic function (the red button) will still operate as before.

When pushed, the panic function instantly turns the speed of all locomotives to zero.
To cancel a Panic situation, press the 'C' key.
It does not cut the power to the track. Accessory 90 will do that (see above).

Errors

If you happen to mis-key any default Zero 1 commands, the yellow Error light will illuminate.

To clear the Error state, press the 'C' key.

Unfortunately, if you mis-key any of the Zero DCC interface commands, they will just not work, or give you un-expected results.

Remember, that Zero DCC interface Accessory functions 99 and 98 can be turned off by typing '99 - ← - ←' and '98 - ← - ←' respectively.

Overload

With the use of the Zero DCC interface, hopefully you will never see the Overload red light illuminate. If you do, it is likely because of a short circuit in the Zero 1 track or accessory power outputs.

Ensure that there are no short circuits and press the 'C' key to clear the message.

If it continues, turn off your Zero 1 and the Zero DCC interface and check all the wiring again.

If the Zero DCC module overloads, track power will be turned off (check LCD display) and power will be cut to all tracks connected to the DCC outputs. If this does not resolve, check for shorts on the track and in your layout wiring.

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Locomotive control

To select a locomotive from the 16 on your list, select 'Loco – *number_to_16* - →'

(in this case you can use either arrow to finish the sequence).

Once you have selected the loco, then the Master Controller slide controller (Speed Regulator) will provide throttle and the FORWARD and REVERSE buttons will control the direction of travel.

Slave controllers

You can have up to three slave controllers by default. This includes one hand held slave controller. Controllers are numbered from the left to right (ie the Master Controller is Slave number 1).

For the more advanced user, with a little bit of modification, you can have up to 15 slave controllers. With the Master Controller that means you can have one controller per locomotive!

To use a slave controller, select:

'Slave – *number_of_slave* – Loco – *number_of_loco* - →'

The selected slave will then have control of the locomotive nominated.

Inertia

Zero 1's in built inertia control will continue to function if desired. NB: This is in addition to any inertia you have pre-programmed into your locomotive's DCC decoder.

Zero 1 has four inertia levels from minimum 1 to maximum 4.

To use Zero 1's inertia, select 'INERTIA – *number_from_1_to_4* – Slave – *number_of_slave* Loco – *number_of_loco* - →'

The selected slave will then have control of the locomotive nominated with the level of inertia selected.

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Point/Turnout control

Due to the new features added to the Zero 1 interface to DCC, the number of points able to be controlled has been reduced to 89. Accessories 90-99 are used for other functions.

If access to a larger number of points or accessories is needed, please contact the provider of the interface for a special build.

To change the state of any powered points 1-89, select:
'*point_number* - →' (or '←')

NOTES:

- This function will only work when Track Power is turned ON (see Accessory 90)
- If the relay is fitted and wired correctly, this function will activate BOTH Zero 1 and DCC points of the same identification number
- The status of your points may be viewed on a correctly configured Hornby Micromimic, and/or a correctly configured e-MicroMimic (see Accessory 93)

Micromimic Console and Displays

If you have the relay switch fitted and correctly wired on your ZeroDCC module, the Zero 1 Phase 3 Micromimic Console and Display should function as normal.

However, to overcome some issues that the Zero DCC interface might cause, rather than connecting it directly to the Zero 1 Master Controller, the Zero DCC interface is supplied with a special output for the Micromimic Console.

This output is switched off by the Zero DCC interface when ever a special accessory function (ie 90-99) is used so that the Micromimic Display continues to show the correct alignment of points while these functions are used. The Zero 1 output is switched back to the Micromimic Console when these functions have ended.

An option is to include Micromimic LEDs for accessories 90-99 placed to indicate the status of these functions on the Micromimic Display Panels (eg to show that Accessory 90 for track power is on or off).

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e-MicroMimic (Accessory 93)

The LCD attached to your ZeroDCC can be configured to act as an ‘electronic’ version of the MicroMimic. It can be used to display a schematic of your layout, including points and signals.

Just like a real MicroMimic, the schematic will display any changes in point or signal directions. Due to the small size of the screen, a Green dot displays the current switched state and a Red dot displays the alternative un-selected state of a point. This is similar for signals.

A special file is required on the SD card in the reader attached to your ZeroDCC. See how to configure it below.

To switch to the e-MicroMimic select Accessory 93:

‘9 - 3 - →’

To switch back to the Locomotive Control Screen, de-select Accessory 93:

‘9 - 3 - ←’

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Configuring your e-MicroMimic Display

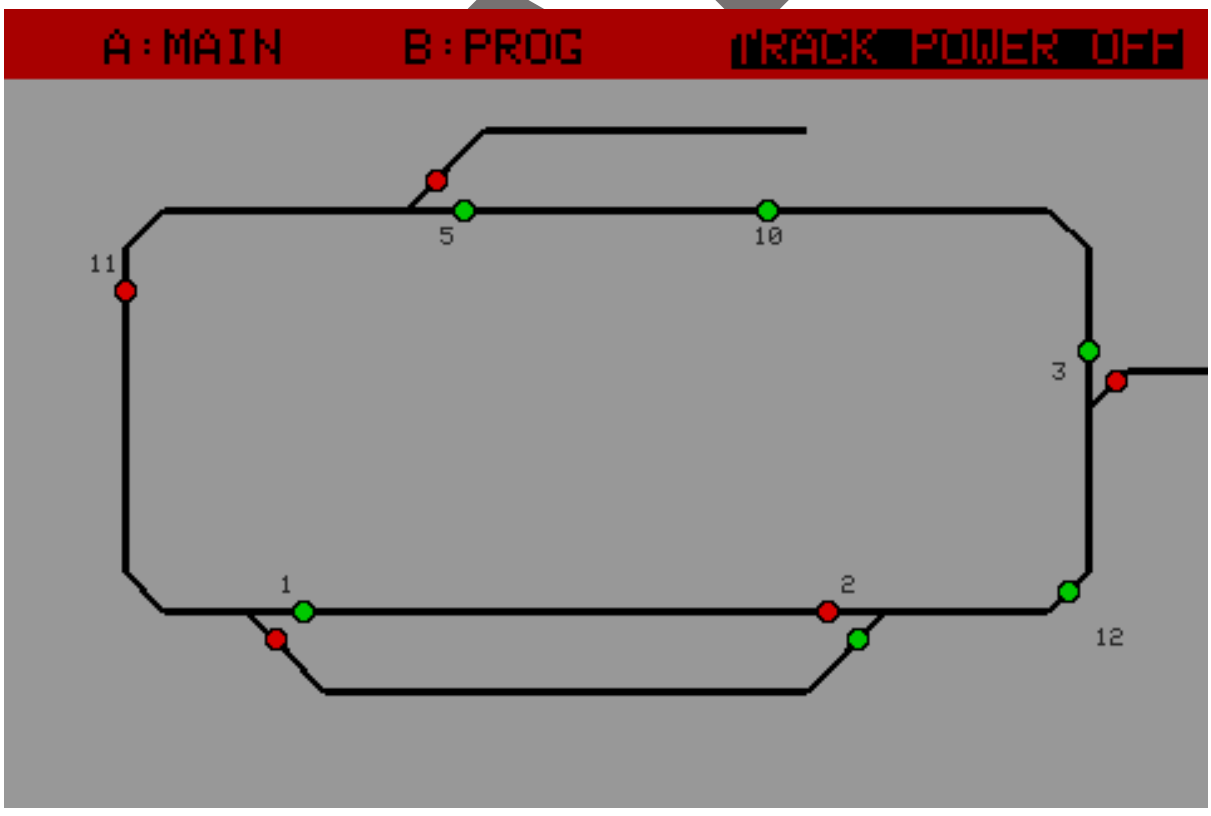
To configure your MicroMimic, a file entitled “Z_DCCMAP.TXT” is required on the SD card in the reader attached to your ZeroDCC.

The file describes what should appear in each square of a 15 column x 9 row matrix.

A sample file is included in the GitHub repository:

```
; Z_DCCMAP.TXT - example layout for ZeroDCC (15 codes per row, 9 rows)
; Oval with a passing loop (points 1 and 2), a siding off the top (point 5)
; a spur off the right-hand side (point 3) and signals 10, 11 and 12.
. . . . . BR H H H H . . . . .
. BR H H H HLT5 H H H HS10 H H H LB .
. VS11A . . . . . . . . . . V .
. V . . . . . . . . . . VBR3 H
. V . . . . . . . . . . V .
. V . . . . . . . . . . V .
. TR H HLB1 H H H H H H HBR2A H H TLS12 .
. . . TR H H H H H H TL . . . .
BLANK
```

Which, when rendered, appears as:



Z_DCCMAP.TXT map file legend

Track



H
Horizontal (mid-left to mid-right)



V
Vertical (middle-top to middle-bottom)



BR
Middle-bottom to mid-right



LB
Mid-left to middle-bottom



TL
Middle-top to mid-left



TR
Mid-right to middle-top

Points - horizontal (12 = accessory number 1-89)



HLT12
Horizontal + mid-left to mid-top



HTR12
Horizontal + mid-top to mid-right



HLB12
Horizontal + mid-left to mid-bottom



HBR12
Horizontal + mid-bottom to mid-right

Points - vertical



VBR12
Vertical + middle-bottom to mid-right



VLB12
Vertical + mid-left to middle-bottom



VTL12
Vertical + middle-top to mid-left



VTR12
Vertical + mid-right to middle-top

Signals - any track code + S + accessory number



HS12
Horizontal with signal



VS12
Vertical with signal



BRS12
Middle-bottom to mid-right with signal



LBS12
Mid-left to middle-bottom with signal



TLS12
Middle-top to mid-left with signal



TRS12
Mid-right to middle-top with signal

Alternative orientation - add A to any point or signal



HLT12A
Same as HLT12, dots reversed



VBR12A
Same as VBR12, dots reversed



HS12A
Same as HS12, dot reversed

Other codes

. or 0	Blank square
BLANK	On its own line: the whole row is empty
; or #	Start of a comment line (ignored)
- and |	Can be used instead of H and V

- Up to 15 codes per row (spaces or commas between them) and up to 9 rows (the top strip shows the status line). Missing codes at the end of a row are bl
- Diagonal letters can be in either order (HLT12 = HTL12). Upper or lower case both work.
- Points: accessory OFF = straight route green, ON = diverging route green. Signals: OFF = green, ON = red. A reverses either.
- Tiles show accessory OFF. The small number is the accessory number (the tiles show the numbers used for this sheet).

To record a Zero 1 session for play back (Accessory 95)

You can record the movement and function control of locomotives and accessories to SD card and play them back at any time.

Note: Unfortunately, this does not include any Zero 1 original accessories or the MicroMimic, but does apply to all DCC functions including lighting and sound controls.

It is important to remember, that the timing will be based on the conditions of your track at the time of the recording. Any change to the maintenance of locomotives, points or the state of the track (including cleaning) will likely affect the timing sequence of the recording or playback. So, make sure your layout is operating correctly before making any recording, or playback.

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Recording

1. Before you start, make sure your locomotives, track and points are all operating efficiently

2. To select the recording function, turn the accessory on, select '9 - 5 - →'

3. Select the number to set the recording to (1-89) and press the right arrow,

   eg '1 - →'

   From now on, any changes made to locomotive control, locomotive functions and point position will be recorded to SD card until you turn the recording off.

4. To end the recording, select '9 - 5 - ←'

Example: You want to make a recording of the locomotive leaving a siding, make a loop of the track and stop at a station:

Make sure track and locomotive are in good condition before you start

9 - 5 - → *enter recording mode*

5 - → *select to make recording number 5*

Use the Zero 1 to open the siding point, take the train out, close the point again, travel around the loop and stop at the station.

9 - 5 - ← *to turn off the recording and exit Accessory 95*

Playback

1. To play back, while still in the recording function (95), select the number of the recording to play back (1-89) and press the left arrow, eg select '1 - ←'.

   During playback the Locomotive Control Screen and e-MicroMimic will display changes as the recording proceeds – even though the Zero 1 locomotive direction and throttle controls have not moved.

   While the play back is in progress, you can still control your locomotives, accessories and DCC functions – this includes those controlled by the recording. Remember that the recording will not adjust for any changes you make, it will continue as originally recorded at the prescribed timeframes.

2. If you need to end the play back early, then select turn off Accessory 95:

   '9 - 5 - ← ' **

   ** Remember if locomotives are moving when you cancel a playback – they will keep moving and you will have to use the physical controls to stop them!

Accessory 95 command functions during recording and playback

Stage	**95->**	**95<-**
Accessory 95 off	Start: wait for a recording number	Ignored (already off)
Waiting for a recording number	Ignored, still waiting	Cancel
Recording	End and save	End and save
Playing back	Stop	Stop

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To change the locomotives in your control list (Accessory 96)

Zero 1 only allows control of 16 locomotives at a time. By default they are set to DCC decoder 1-16. However, you can change them at any time to any locomotive in your roster. Once changed, they are automatically stored on a backup SD card and restored on your next session.

Zero 1 does not allow the use of zero (0) as an accessory. If a zero is needed in your DCC command sequence, use 10 as a work around (see example below.)

1. To turn the list control accessory on, select '9 - 6 - →'

   To turn the list control accessory off, select '9 - 6 - ←'

2. To change a locomotive in your list:
   a. Enter the DCC CAB ID one digit at a time with a '←' to confirm each digit
      NB: to enter a 0, use 10 instead!
   b. To add this locomotive to the control list, enter the desired number from the control list (1-16) and confirm with →'

3. You can change as many locomotives in the control list as you like while the accessory is turned on!

4. To turn the accessory off, select '9 - 6 - ←'

   Example: If you wanted to add a locomotive with the DCC CAD ID of 909 to Zero 1 control list number 14 (ie locomotive number 14) the sequence would be:

9 - 6 - →	*to turn on the accessory*
9 - ← - 10 - ← - 9 - ←	*to enter the DCC CAD ID (remember 0's are 10)*
14 - →	*to enter the Zero 1 control list number to assign*
	you can change as many as you like, then
9 - 6 - ←	*turn off the accessory*

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To set up a new list of locomotives, or recall a previous one (Accessory 97)

Accessory 96 (above) allows you to set which DCC locomotive ID is set up in Zero 1's list of controllable locomotives, and saves it on SD card.

Accessory 97 allows you to set up multiple lists of 16 locomotives.

So, you could have a list of:

- 16 favourite locomotives
- 16 least favourite locomotives
- 16 steam locomotives
- 16 diesel locomotives
- 16 locomotives from the 1920's
- 16 locomotives from the 2020's
- And so on….

To use this function, remember that when changing a locomotive in Accessory 96 (above), that the change is automatically changed in the current list.

If you haven't created a new list, Accessory 97 then takes the current list and assigns it a new number (1-10), then any changes made in Accessory 96 are recorded in the new list.

If you have already created a new list, Accessory 97 will load it from the SD card for you.

1. To turn the list control accessory on, select '9 - 7 - →'

2. To change a locomotive in your list, then enter the list number you wish to create or recall (1-10) and confirm with →'

3. To turn the accessory off, select '9 - 7 - ←'

   Example: If you wanted to add a new list numbered 3:

9 - 7 - →	*to turn on the accessory*
3 - →	*to enter the list number to assign*
9 - 7 - ←	*turn off the accessory (technically not needed as this accessory will turn itself off after a list is chosen)*

   Remember, if you already have a list '3' created this will load it for you. If you don't have a list '3' this will create it based on the current locomotive roster in place and will be saved after you make the first change using Accessory 96 (above).

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Head light and whistle/horn control (Accessory 98)

NB: You must have a sound equipped and programmed DCC locomotive for the whistle/horn to work, and a headlight equipped and programmed DCC locomotive for the headlight to work!

1. To turn the head light and whistle/horn accessory on, select '9 - 8 - →'

   To turn the head light and whistle/horn accessory off, select '9 - 8 - ←'

2. Once the function is turned on, select the number of the locomotive

3. To turn the head light on and off, select '→'

4. To turn the whistle/horn on and off, select '←' (NB: remember it will keep going until you toggle it back off!)

While the locomotive is selected you can keep toggling the head light and whistle/horn on/off as many times as you like.

While the accessory is on, you can change the number of the locomotive at any time.

To turn the function off, select '9 - 8 - ←'

Example: If you wanted to blow the whistle on a sound and light equipped DCC locomotive number 12 and turn on the headlight the sequence would be:

9 - 8 - →	*to turn on the function*
12	*to enter the locomotive number*
→	*to blow the whistle*
→	*to make the whistle stop!*
←	*to turn on the headlight*
	you can blow the whistle, and turn on and off the headlight as many times as you like, until you change the locomotive number or turn off the accessory
	you can change loco numbers as many time as you like, until you turn off the accessory
9 - 8 - ←	*turn off the accessory*

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DCC CAB function controls (Accessory 99)

NB: You must have a decoder programmed for multiple functions for this to work!

1. To turn the CAB functions accessory on, select '9 - 9 - →'

   To turn the CAB functions accessory off, select '9 - 9 - ←'

2. Once the accessory is turned on, select the number of the locomotive

3. Next confirm with '→'

4. Next select the number of the CAB function (eg whistle = 2)

5. To turn the function on and off, select '→'

The locomotive will remain selected (until you turn Accessory 98 off). While the current function is selected you can keep toggling it on/off as many times as you like.

While the accessory is on, you can change the function number at any time.

To turn the function off, select '9 - 9 - ←'

Example: If you wanted to blow the whistle on a sound and light equipped DCC locomotive number 12 the sequence would be:

9 - 9 - →	*to turn on the function*
12 - →	*to enter the locomotive number*
2 - →	*to toggle function 2 (ie blow the whistle)*
→	*to toggle it again*
	you can change between functions on the selected locomotive as many times as you like, until you turn off the accessory
9 - 9 - ←	*turn off the accessory*

TIP: If your DCC decoder requires you to 'start up' the locomotive with CAB Function 8, use this Zero DCC accessory function (just replace the '2' in the above example with an '8'!)

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JMRI: A Java Model Railroad Interface

The Zero DCC interface also provides a USB port to connect to your computer to provide an interface with the open source JMRI applications and associated functions.

But it's much more fun controlling your trains with the Zero 1!

Further JMRI information and downloads are available at: https://www.jmri.org/

WiFi access for computer or mobile device

The Zero DCC interface is also capable of WiFi Connections for JMRI WiThrottle compatible applications (eg Android Engine Driver or iOS WiThrottle).

The Zero DCC interface will operate as an independent WiFi access point. On your device, connect to the access point with the name commencing with "DCC" the remainder of the name is linked to your individual interface's network MAC address.

The password for your device is "Password" plus the MAC address that was part of your interface's identifier.

Once you are connected to your interface's local WiFi network, you can use your JRMI WiThrottle application to control your locomotives and points.

In your mobile application, you will need to point it to the following:

IP address: 192.168.0.4
Port: 2560

Remember, you are accessing the DCC controller directly, and can only control your DCC locomotives and any DCC wired points – you cannot change your Zero 1 points or accessories using the WiThrottle; and your Micromimic Display will not display any point changes.

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To program the DCC decoder ID (CV1) on a separate program track (Accessory 91)

Warning: This feature of the Zero DCC interface comes with some risk!

Power interruptions, timing issues and other unforeseen problems may damage your decoder. Consider having your locomotive programmed by a professional.

USE THIS FEATURE AT YOUR OWN RISK – YOU HAVE BEEN WARNED!!!!

If you choose to proceed, please be very careful and double check everything you are planning to do.

You cannot do this on the main track. You will need a separate programming track connected to the programming track outputs on the Zero DCC interface.

Consider the DCC locomotive ID you wish to use. Some things to consider:

- All decoders come with a default DCC ID of 3. So best to avoid that.
- Using a DCC ID of zero (0) can cause problems with your configuration – so the Zero DCC interface has been written to prevent you doing this.
- There are a number of DCC systems that will not recognise a DCC ID over 99 – so the Zero DCC interface has been written to prevent you doing this (so that if you take your locomotives to another layout using one of these systems it will still work).

Please read the following instructions several times and work through what you will be doing BEFORE attempting this RISKY procedure.

Remember on the Zero 1 you will not have any acknowledgement that the key press was recognised.

You will need to be planned and organised; slow and deliberate.

Write out your key press sequence in advance to help prevent mistakes.

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READ THIS SEVERAL TIMES:

1. Prepare for the process and ensure that power, track connections or any other issues that will interfere with the process are prevented.

2. Set up your programming track.

   - It cannot be connected to the main line in any way.
   - The locomotive may move during programming, so ensure the track is in a safe location. Consider buffers at each end.
   - Connect the programming track to the programming track outlets on the Zero DCC interface.

3. DO NOT PUT YOUR LOCOMOTIVE ON THE PROGRAMING TRACK YET

4. Turn the power to the track OFF, select:

   '9 - 0 - →' *to enter Accessory 90*, then
   '2 - →' *to set the DCC Main and Prog channels and turn power off,* then
   '9 - 0 - ←' *to leave Accessory 90*

5. Confirm that the track power is off

6. To be safe, remove all locomotives from all tracks.

7. Now, place the locomotive you wish to program onto the separate programming track.

8. Turn the power to the track on, select

   '9 - 0 - →' *to enter Accessory 90*, then
   '1 - →' *to select power on,* then
   '9 - 0 - ←' *to leave Accessory 90 and turn the track power on.*

9. Turn on the programming accessory on, select '9 - 1 - →'

10. Now carefully enter the new DCC ID you want to program:

    a. Enter the DCC CAB ID one digit at a time
    b. Confirm each digit with a '←'.

    NB: to enter a zero (0), use 1-0 instead!

11. Up to this point, if you make any mistakes you can cancel the entire programming routine by selecting '9 - 1 - ←'

12. When you have finished carefully entering the new DCC ID, to commence the programming sequence, press the '→' key

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13. Remember, the locomotive may move while being programmed so be prepared. DO NOT ALLOW IT TO LEAVE THE TRACK!

14. As the Zero 1 has no feedback mechanism to advise you that this was completed, wait a few minutes before removing the locomotive. You may hear some relays click in the Zero DCC interface when the process has completed, but still wait a few minutes before removing the locomotive.

15. You do not need to end the programming routine, it will end automatically after the programming commands have been sent to the DCC base station (or if you key any incorrect sequences. Eg using a DCC CAB ID of zero.) But you can still turn off the programming routine by selecting '9 - 1 - ←' to confirm. If you are using a Micromimic Display and have an LED set up for accessory 96 you will see it change – that is an indication that the DCC programming is complete.

16. Test your locomotive on the main line. It should now respond to the new DCC CAB ID – remember to add it to your roster (Accessory 96) before you try testing it.

17. Read all of this again, and write down your key sequence.

18. Read it again.

Problem solving the programming function.

If your locomotive does not respond to its new DCC CAB ID, here are some things to consider. You will see a theme here. This process is risky using the Zero 1. Consider having your locomotive programmed by a professional.

- Did you try selecting a DCC CAB ID of zero (0) or over 99? The software deliberately prevented you from programming these. Read the manual again. Consider having your locomotive programmed by a professional.

- Does the locomotive respond to its old DCC CAB ID (most are default 3 when new)? The process didn't work. Read the manual again. Consider having your locomotive programmed by a professional.

- Could you have mis-keyed during the process? Read the manual again. Consider having your locomotive programmed by a professional.

- Did the power fail during the process? This could have damaged the decoder. Consider having your locomotive programmed by a professional.

- Did the locomotive loose contact with the programming track at any point during the process? Read the manual again. Consider having your locomotive programmed by a professional.

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- Did any of the wiring connections come loose at any point during the process? Read the manual again. Consider having your locomotive programmed by a professional.

- Did you try programming the locomotive on the main track? Read the manual again. Consider having your locomotive programmed by a professional.

- Consider having your locomotive programmed by a professional.

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Acknowledgements

We would like to thank Hornby Hobbies and their forebears for creating the Zero 1 control system. It was ahead of its time, and hopefully this project will demonstrate that the system is still capable in the 21st century.

We would also like to thank all the developers of the DCC++EX (https://dcc-ex.com) open source base station software and hardware schemes, on which this interface is based, for making such a flexible and adaptable platform for digital model railroading.

Thanks also to Geoff Bunza for his excellent, simple DCC sniffer circuit and PCB design.

We make no claims for the excellent work of all these people over such a long time. We are indebted to you all.

We do, however, take great credit ;) for creating the interface that allows these two great systems to talk to each other and hopefully keep the Zero 1 community alive and creative in the 21st century.

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